Vectorized Streaming Data Pipeline: DuckDB, Apache Arrow & In-Memory LRU Subgraph Cache

1. Omni-Channel Ingress

SWIFT MT103 / ISO-20022
Cross-Border Core Wires

Bitcoin & EVM Mappings
UTXO / ERC-20 Event Logs

Fast P2P / Mobile Money
bKash / ACH Micro-Transfers

Streaming Ingest Buffer
Kafka / Redpanda: 31,200 msg/s

2. Causal Dual-Path Engine

FAST-PATH: In-Memory LRU Cache
88.4% Hit Rate | 100k Ego-Graphs
Sub-Millisecond Hit SLA: 0.45 ms
Zero SQL Overhead | Thread-Safe RLock

Populate

SLOW-PATH: Embedded DuckDB OLAP
11.6% Cache Misses | Cold Ego Scan
Composite B-Tree: (u, t_epoch DESC)
Zero-Lookahead Filter: t_e <= t (2.1 ms)
Materializes v_canonical_invariants_12d

3. Apache Arrow C-Interface

Shared RAM Virtual Memory
DuckDB RecordBatch Chunk
Pointer Handshake via C ABI
Zero-Copy Tensor Memory Pointer

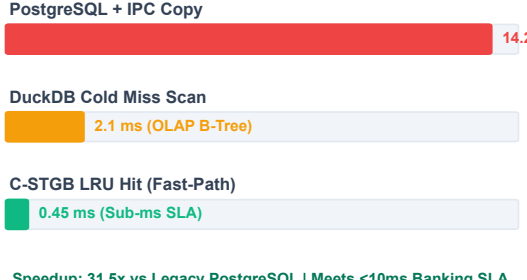
44x Ingestion Speedup
14.2 ms (Copy) -> 0.32 ms
0 Heap Copies | No Serialization

4. PyTorch Geometric

HeteroData Causal Graph
Node Tensors: X_u in R^(N x d)
Edge Index: E in R^(2 x M)
12-D Invariants: z_inv in R^12

To Spatio-Temporal GNN
3-Band Harmonic Attention
Continuous Message Passing

(a) End-to-End Ingestion Latency (ms)



Speedup: 31.5x vs Legacy PostgreSQL | Meets <10ms Banking SLA

(b) Memory Copy Overhead & Data Interface

Legacy Pipeline: 3 Copies / Serialization

SQL Query → JSON/Proto → Socket IPC → PyG Tensor

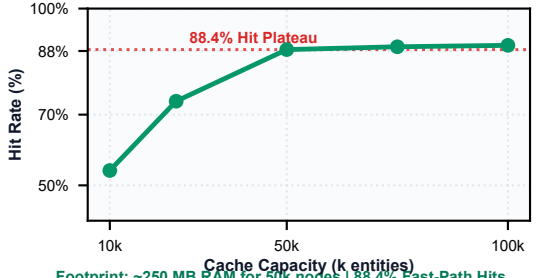
Latency: 14.2 ms / 10k edges | 3 Heap Re-allocations

C-STGB Arrow C-Interface: Zero-Copy Handshake

Contiguous Shared Memory (Arrow RecordBatch)
Pointer Pass via PyCapsule -> Instant PyG Tensor (0.32 ms)

Memory Benefit: 0 Heap Duplication | 44x Tensor Ingest Speedup

(c) In-Memory LRU Cache Hit Dynamics



Footprint: ~250 MB RAM for 50k nodes | 88.4% Fast-Path Hits